COAGENTICS: EXPERIMENTAL IDENTIFICATION AND VALIDATION OF AI BEHAVIOR IN DYNAMIC ECONOMIC MARKETS

Five-Page Technical White Paper | Influence Benchmarks for AI Systems

INVIA Digital | Principal Investigator: Paul Cummings, Ph.D.

1. Technical Problem and Research Objective

AI agents are beginning to act inside markets, negotiations, resource-allocation systems, and other environments in which their outputs change the state subsequently observed by other agents. Static benchmarks do not capture this feedback. A model can produce a plausible answer in isolation yet behave differently when price history, incentives, peer actions, or information reliability change.

CoAgentics treats that problem as experimental science. The system represents a controlled domain, competing predictive models of observable behavior, interventions, validation rules, and evidence. For DV026, the research extends this foundation into a multi-agent economic laboratory in which black-box LLM agents submit economically consequential actions through a shared market mechanism.

Research hypothesis

The primary falsifiable claim is that adaptive experimental selection can identify or reject competing predictive models with fewer LLM queries than fixed-factorial or randomized designs, without weakening calibration, held-out prediction, or independent replication. When the evidence cannot distinguish models, CoAgentics reports non-identifiability rather than assigning a behavioral label.

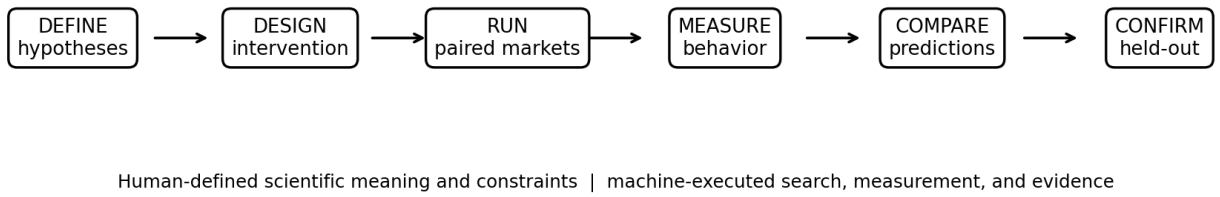


Figure 1. CoAgentics experimental loop. Researchers define hypotheses and constraints; the system executes controlled tests, measures observable behavior, and retains evidence.

Why economic markets

Experimental markets convert language-model behavior into commitments: bids, offers, quantities, timing, revisions, trades, and realized surplus. Induced values and costs create a known efficient allocation, allowing the experimenter to compare observed outcomes with a calculable reference. Continuous double auctions are particularly useful because individual actions and collective price formation are both observable. Smith's foundational experiments established the double auction as a rigorous setting for testing competitive-market behavior; recent LLM-market studies show that model families can differ materially in convergence, efficiency, and strategic behavior. [1-4]

2. Experimental Design: Make Competing Models Disagree

The research does not ask an LLM whether it is biased, influenced, collusive, or rational. It specifies competing observable-response models and constructs interventions under which those models predict different actions. The strongest experiment is the feasible condition that maximizes predicted separation among currently plausible models, subject to query budget and researcher-approved market semantics.

Market environment

Each market assigns buyers private values and sellers private costs. Agents observe a controlled public state - best bid/ask, last trade, permitted history, and experimental information - plus their own role and reservation value. They return BUY, SELL, or HOLD with price and quantity. Invalid actions are recorded as invalid rather than silently repaired. Phase I uses a continuous double auction as the primary mechanism and a second mechanism for held-out generalization.

Controlled variables and observable responses

Experimental factor	Example intervention	Measured response
Information	truth status, relevance, reliability, timing, wording	bid/offer shift, participation, latency
Market history	rising/falling prices, spread, recent trades	shading, revisions, price dependence
Peers	visible actions, population/model composition	peer dependence, propagation, collective outcome
Incentives	private values/costs, payoff structure	reservation-value deviation, surplus allocation
Mechanism	CDA vs. held-out mechanism	generalization, efficiency, price formation

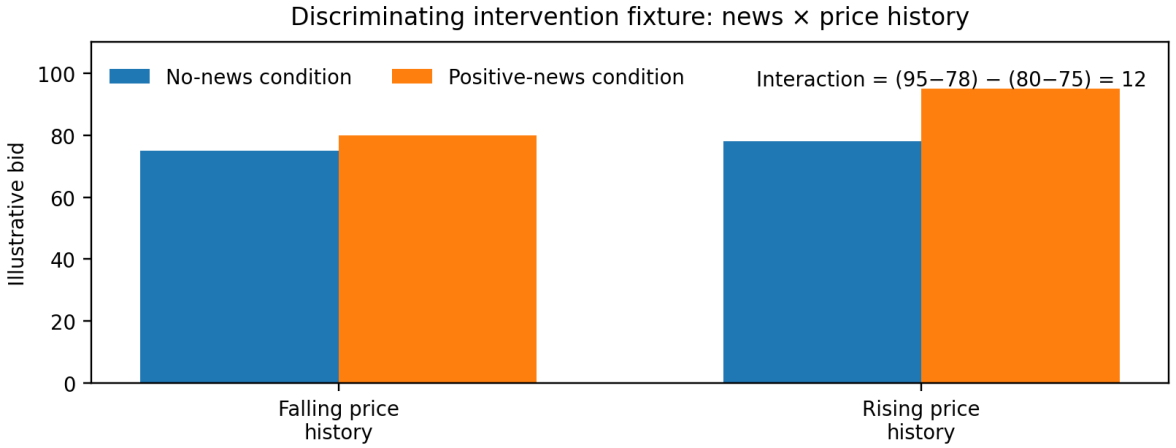


Figure 2. Illustrative software fixture. A 2×2 news × price-history design creates a measurable interaction of 12 bid units; it is a test fixture, not an empirical LLM finding.

Statistical discrimination

For a two-model comparison, the selector can rank candidate interventions by predicted behavioral separation per unit experiment cost. With more than two models, expected information gain provides a natural extension. Discovery data may eliminate weak explanations or expose structured residuals. Any revised hypothesis created from discovery evidence must face fresh confirmation data. Confirmation conditions, stopping rules, exclusions, sample sizes, and failure handling are frozen before the held-out test.

3. Analytics: From Trades to Behavioral Evidence

CoAgentics analyzes the market at three levels: action, agent, and system. Action-level measures capture bids, offers, holds, revisions, latency, and validity. Agent-level measures estimate treatment response, sensitivity to information reliability, price-history dependence, peer-action dependence, and persistent deviation from reservation-value behavior. System-level measures capture allocation, price convergence, spread, volume, gains from trade, and the distribution of surplus.

Core market metric

Allocative efficiency is the fraction of maximum feasible gains from trade actually realized: $\eta = G_{realized} / G_{optimal} \times 100\%$. Because private values and costs are assigned by the experiment, $G_{optimal}$ is known before trading begins. Efficiency is necessary but not sufficient: two markets can have similar efficiency while exhibiting different prices, bidding distributions, convergence, or surplus allocation.

Preliminary engineering evidence

A recent engineering feasibility run exercised 10 distinct LLMs through 14 model calls and recorded three independently logged LLM-to-LLM trades. Realized surplus was 50 against 110 maximum feasible surplus, or 45.45% allocative efficiency. These observations demonstrate an end-to-end path from model call to market action to recorded trade. They are not an independently replicated scientific result and do not satisfy the DV026 market-qualification target.

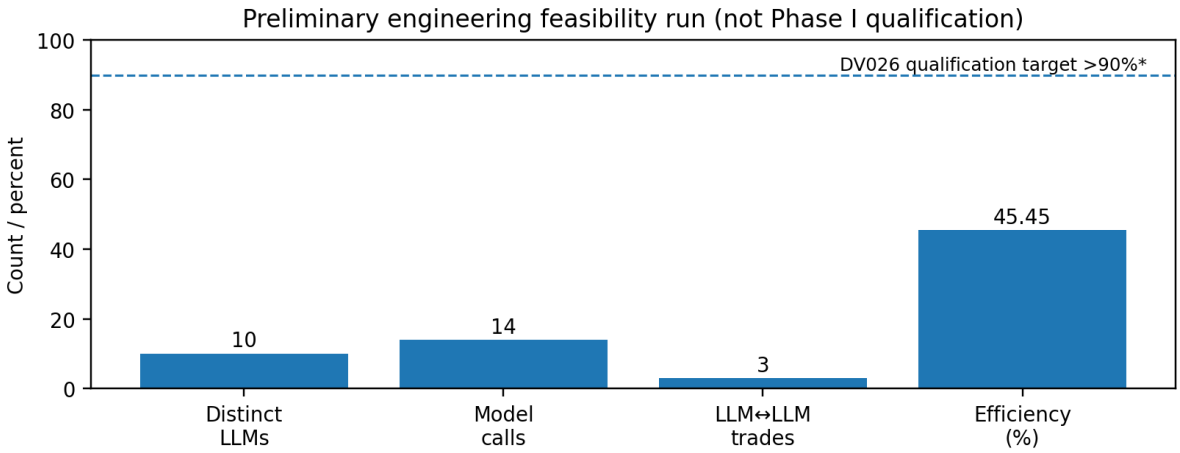


Figure 3. Preliminary feasibility analytics. The run demonstrates technical execution across 10 LLMs; scientific validation and >90% qualified reference-market efficiency remain Phase I work. *Under documented qualification conditions.

Paired counterfactual analysis

The principal causal unit is a matched market pair: hold seed, private values, agent identities, and mechanism constant; change one controlled treatment. For treatment z, estimate the paired effect $\Delta = Y(z=1) - Y(z=0)$ for outcomes such as bid shift, participation, latency, efficiency, or surplus. Repetition across independent seeds yields an effect distribution rather than a single anecdote. Factorial designs test interactions, such as whether the response to news depends on price history or peer behavior.

Model-quality analytics

Candidate models are scored on held-out prediction, calibration, reproducibility, and generalization. The system retains uncertainty and abstains when evidence is insufficient. This separates an observed treatment effect from an interpretation of internal mechanism: randomized interventions support claims about observable response, not access to an LLM's hidden beliefs or reasoning.

4. Architecture, Evidence, and Reproducibility

The experimental system separates the market engine from model providers. A provider adapter freezes model identity/configuration and exposes a standardized action protocol. The market engine owns state, matching, settlement, and experimental timing. This prevents provider-specific behavior from being confused with market logic and allows the same experiment manifest to be replayed across model families.

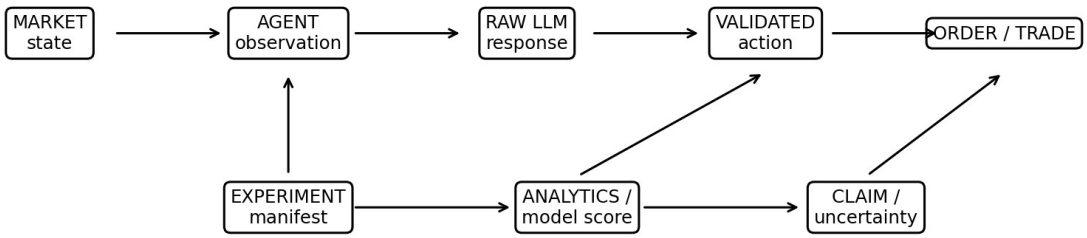


Figure 4. Evidence lineage. Each claim can be traced from experiment manifest and market state through the exact observation, model response, validated action, order/trade, analytics, and uncertainty.

Evidence record

For each experimental unit, the retained record includes campaign and experiment identifiers; discovery/confirmation status; seed; model/provider configuration; treatment assignment; exact agent observation; raw model response where policy permits; parsed and validated action; latency; submitted order; trades; account changes; calculated metrics; analysis version; and claim status. Provider failures and invalid outputs are recorded separately from behavioral outcomes.

Qualification sequence

- Programmed-agent ground truth: verify matching, settlement, efficient allocation, and known-rule behavioral classifiers before interpreting LLM behavior.
- Reference-market qualification: demonstrate >90% allocative efficiency under documented qualification conditions while also checking transaction prices, bid/offer distributions, convergence, volume, and surplus allocation.
- Cross-model execution: exercise stock agents from at least 10 distinct underlying LLMs in actual markets; configuration variants, replay agents, and mocks do not count as distinct models.
- Discovery and discrimination: run controlled interventions, compare predictive models, quantify uncertainty, and identify non-identifiable regions.
- Held-out confirmation: freeze models and decision rules, test reserved conditions and a second market mechanism, and reproduce results from retained manifests.

Human reference

George Mason University supports market validity, behavioral measures, experimental discriminators, and comparison with established experimental-economics results and calibrated computational models. Phase I can use existing human-market evidence and reference models; it does not require new human-subject experiments. The comparison asks whether AI markets reproduce, depart from, or systematically transform known market regularities.

5. Research Plan, Expected Results

Phase I technical objectives

Gate	Technical objective	Evidence	Decision
M2	Qualified market foundation	programmed-agent tests; settlement; evidence schema	market mechanics accepted/reworked
M6	Experimental protocol + analytics	paired/factorial interventions; model scoring; selector baseline	discrimination measurable or redesign
M9	≥10 distinct LLMs exercised	identity/config manifests; actions; trades; outcomes	cross-model campaign executable
M12	Qualification + held-out confirmation	>90% reference efficiency; behavioral fidelity; human-reference comparison; reproducibility	feasibility supported, bounded, or rejected

Expected research products

Phase I will produce an executable multi-agent market benchmark, a documented stock-agent interface spanning at least ten distinct LLMs, market-qualification tests, behavioral-response classifiers tied to operational definitions, adaptive and fixed/random experiment-selection baselines, held-out confirmation results, human-reference comparisons, and an auditable evidence package. A negative result is scientifically valid: failure to achieve the efficiency gate, failure of adaptive selection to reduce query cost, or persistent non-identifiability will be reported directly.

Technical significance

The differentiator is controlled model discrimination, not another leaderboard. A leaderboard ranks models under a fixed task. CoAgentics asks which explanation predicts an agent's behavior, chooses conditions that separate the explanations, executes the experiment, and then tests the surviving explanation on data that were unavailable during discovery. In multi-agent markets, the same method can determine whether individually characterized responses remain predictive after agents interact and whether collective outcomes require interaction-specific terms.

Transition

The economic benchmark provides a compact, measurable proving ground for a broader AI test-and-evaluation capability. Government applications include AI T&E, autonomy assurance, acquisition evaluation, mission-system validation, wargaming, logistics, and decision support. Commercial applications include AI-agent evaluation, model assurance, economic simulation, and integration with customer simulation/test environments. The reusable product is the experiment-validation-evidence architecture; the market benchmark is the first qualified application.

Selected research basis

[1] V. L. Smith, “An Experimental Study of Competitive Market Behavior,” Journal of Political Economy, 1962.
[2] P. Struski et al., “Competitive Market Behavior of LLMs,” arXiv:2609.02580, 2026.
[3] K. Agrawal et al., “Evaluating LLM Agent Collusion in Double Auctions,” arXiv:2507.01413, 2025.
[4] R. M. del Rio-Chanona, M. Pangallo, and C. Hommes, “Can Generative AI agents behave like humans? Evidence from laboratory market experiments,” arXiv:2505.07457, 2025.
[5] J. Piao et al., “MarketSim: Simulating Stock Markets with Large-Scale Generative Agents,” ICML/PMLR, 2026.